

AST 6007-01 Spacecraft Dynamics and Control

Homework #3

Due Nov 10 2021, 10:00AM

In the following problems, we are interested in studying the motion of a rigid spacecraft corresponding to the indicated assumptions. You can use analytical and computer based approaches to answer the posed questions.

1. A rigid spacecraft rotates with a constant angular velocity vector whose components are given by $(0.1, -0.05, 0.03)$ *rad/sec* when expressed in the body fixed coordinate frame. Initially, the body fixed frame of the spacecraft is exactly aligned with the inertial coordinate frame.
 - a. After 10 seconds, what is the attitude of the spacecraft in terms of (i) rotation matrix, (ii) exponential coordinates, (iii) Euler angles, and (iv) quaternions?
 - b. After 10 seconds, what is the angular velocity vector of the spacecraft expressed in terms of the (i) body fixed coordinate frame and (ii) the inertial coordinate frame?
2. A rigid spacecraft has principal moments of inertia $(2000, 2000, 1000)$ *kg-m²* with no external torque acting on the spacecraft. The third principal axis is an axis of mass symmetry of the spacecraft.
 - a. What are all equilibrium solutions of the Euler equations?
 - b. What is the time response for the spacecraft angular velocity vector, expressed in the spacecraft body fixed coordinate frame, corresponding to the initial condition of $(0.0, 0.0, 0.8)$ *rad/sec* expressed in the body fixed coordinate frame? What is the time response for the spacecraft attitude assuming that, initially, the spacecraft body fixed coordinate frame is aligned with the inertial coordinate frame?
 - c. Repeat part b for the initial condition of $(0.05, -0.05, 0.0)$ *rad/sec* expressed in the body fixed coordinate frame.

Provide graphs of these time responses over whatever time interval you think demonstrates interesting response properties, e.g., periodicity or asymptotic convergence. You may use any suitable choice of coordinates to describe the spacecraft attitude responses.

3. Repeat Problem 2 for an *asymmetric* rigid spacecraft with principal moments of inertia $(2000, 1500, 1000)$ *kg-m²*.
4. (Optional Extra Credit Problems) Develop computer animations of the spacecraft motions in the above problems, so that the spacecraft velocity vector and the spacecraft attitude can be visualized. You may use Matlab, Mathematica, or any other animation tools.